

Batch - 05

Title: Scientific Research Paper Recommendation Using Sentence Embeddings and Semantic Search

Abstract

The exponential growth of scientific publications has made it increasingly challenging for researchers, students, and academicians to discover relevant research papers using traditional keyword-based search systems. Conventional search methods often fail to capture the semantic meaning of user queries, resulting in less accurate and relevant recommendations. This project proposes a **Scientific Research Paper Recommendation System** that utilizes **Sentence-BERT (SBERT)** to generate semantic embeddings of research paper titles, abstracts, and keywords, enabling meaning-based retrieval instead of simple keyword matching.

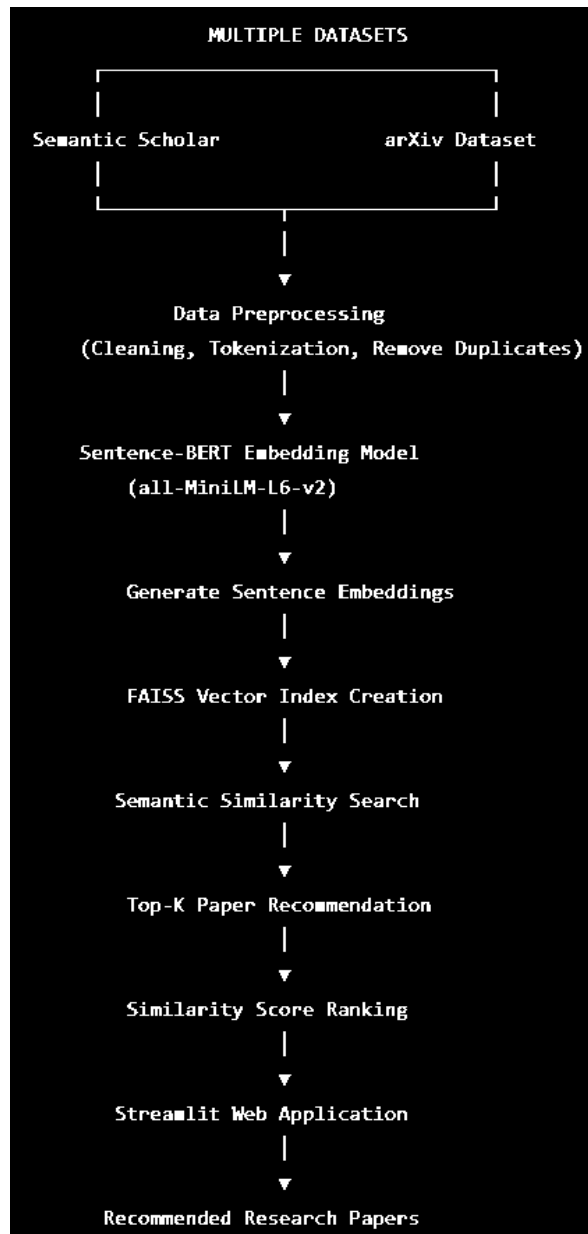
The proposed system preprocesses research paper metadata by cleaning and combining textual information before converting it into dense vector embeddings using the pre-trained **Sentence-BERT (all-MiniLM-L6-v2)** model. These embeddings are indexed and stored in **FAISS (Facebook AI Similarity Search)**, which enables efficient semantic similarity search over large collections of research papers. When a user submits a search query, the system converts the query into an embedding, performs semantic search using FAISS, and retrieves the most relevant research papers based on cosine similarity. The recommended papers are displayed along with their title, abstract, authors, publication year, similarity score, and reference link through an interactive **Streamlit** web application.

The performance of the proposed recommendation system is evaluated using retrieval metrics such as **Precision@K**, **Recall@K**, **Mean Reciprocal Rank (MRR)**, and search response time. The resulting system aims to provide accurate, fast, and context-aware research paper recommendations, assisting users in efficiently discovering relevant scientific literature while demonstrating the practical application of Natural Language Processing (NLP), sentence embeddings, semantic search, and vector indexing techniques.

Datasets

Dataset	URL link
arXiv	https://www.kaggle.com/datasets/Cornell-University/arxiv?utm
Semantic Scholar Open Research Corpus (S2ORC)	https://www.semanticscholar.org/product/api

Approach



Project Group Members

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